

1a) numBeads playerName totalScore team1	1b) 2players — cannot start with a number. player-name — hyphens are not allowed. class — reserved Java keyword. my name — spaces are not allowed.
2a) int numBeads; numBeads = 5;	2b) int numBeads = 5;
3a) yourNumber = 15	3b) yourNumber = 11
4a) int b) double c) int d) double e) boolean f) char	5a) A primitive data type stores a simple value directly, while an abstract data type represents a collection of data and operations.
5b) A class is a blueprint for creating objects. An object is an instance of a class.	6a) Three possible methods: getPlayerCount addPlayer removePlayer
6) Three possible objects: flames teamCanada crescentCowboys	7)import java.util.Random;
8a) $5 + 7 - 3 = 9$ b) $10 * 2 - 3 = 17$ c) $10 * (2 - 3) = -10$ d) $8 - 3 * 2 = 2$ e) $10 / 5 * 4 = 8$ f) $10 / 2 + 3 = 8$ g) $6 \% 3 + 4 = 4$ h) $12 \% 5 * 3 = 6$ i) $12 \% (5 * 3) = 12$	9 $x = 2005$ ans 0 $x = 1776$ ans 7 $x = 39$ ans 3
11a) $y = (int)(j * k)$; 8 after casting/truncating. b) $z = j * k$; No casting needed because the result is automatically a double. c) $z = k * k$; No casting needed because both values are double. d) $j = (int)k$; 1 after casting/truncating. e) $k = j$; No casting needed because an int can automatically be converted to a double. f) $y = j + 3$; No casting needed because both	12 / performs division, while % gives the remainder. Type casting changes the data type of a value and can remove the decimal portion when converting from double to int.

values are int	
13a) total += 10; b) numStones -= 1; c) days %= 24; d) price *= 1.2;	14a) Variable: int votesReceived; b) Variable: double percentageWon; c) Constant: final char FIRST_INITIAL; final char MIDDLE_INITIAL; final char LAST_INITIAL; d) Variable: int electionYear;
15a) Syntax error — duple should be double. b) Syntax error — missing semicolon: int numHats; c) Syntax error — == is a comparison operator, and there is no assignment or complete statement. d) Logic error — division happens before addition, so it calculates test1 + (test2 / 2) instead of the average. It should be: avg = (test1 + test2) / 2; e) Runtime error — division by zero occurs. f) Syntax error — the closing parenthesis is missing: System.out.println(money.format(payCheck));	16a) False — identifiers cannot contain spaces. b) True — double is a floating-point data type. c) False — an abstract data type is different from a primitive data type. d) False — Values typed by the user can be used in an application. e) True — The Scanner class is part of the java.util package. f) False — + and – have the same precedence. g) True — The – operator has lower precedence than %. h) True — byte is a Java keyword. i) False — Java is case-sensitive, so apple and Apple are different identifiers. j) False — Errors that violate Java's rules are called syntax errors, not semantic errors.